# REVEAL: Agentic In-Hand Manipulation for Language-Guided Target Presentation

Anonymous CVPR submission

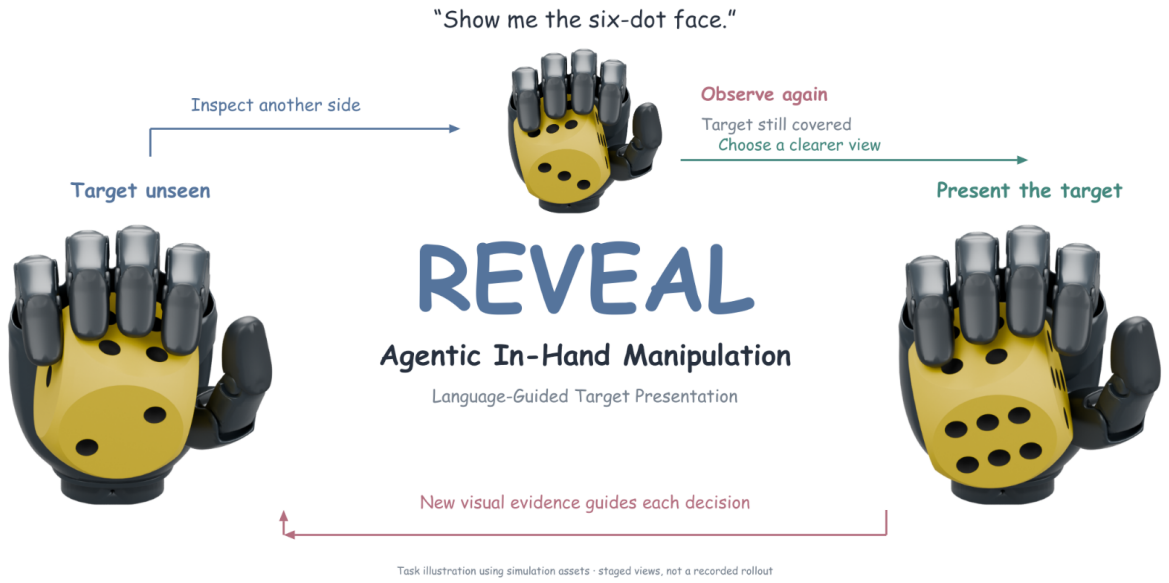



Figure 1. **REVEAL overview.** New visual evidence guides semantic decisions as the hand inspects and presents a requested object face. This staged simulation illustration is not a recorded rollout or physical-robot result.

## Abstract

*Language-guided in-hand manipulation requires a robot to uncover a requested object face while maintaining contact. This couples visual information gathering with motor control, yet high-level visual reasoning introduces delays during which the hand must continue acting. We present REVEAL, an asynchronous framework for online visual search and semantic target presentation. The agent uses updated images to inspect the object and bind the requested target to a shared goal bank, which interfaces with a history-conditioned motor policy. Teacher-supervised local controllers adapt joint commands during pending requests and subsequent execution. Simulation evaluations cover six object-specific systems, with additional English/Chinese instruction tests on three objects. Under changed semantic mappings, REVEAL achieves an object-averaged success rate of 65.8%, exceeding the strongest evaluated baseline by 40.4 percentage points.*

## 1. Introduction

A command such as "present the face marked T" specifies a visual goal, not a sequence of hand motions. The robot must find the mark, associate it with a controllable orientation, and rotate the object while preserving contact. The target may be hidden by the object or the fingers, so an informative action also changes the grasp and the next observation. In-hand target presentation therefore couples visual reasoning with contact-rich control.

Vision-language agents offer a flexible way to interpret such requests, but their decisions arrive asynchronously. During inference, the hand and object continue to move. Repeating the previous command may lose the grasp or leave the target occluded; slowing motion alone may preserve contact without producing a useful correction. A practical system needs a local controller that responds to current measurements while semantic reasoning is pending.

Our key insight is to separate the rate of semantic decisions from the rate of motor control while keeping both interfaces grounded in the same goal representation. An

asynchronous visual agent inspects the object and issues a validated semantic command. A history-conditioned student maps the active goal and measured pose to joint targets. For the cube, separate local controllers adapt those targets during pending requests and ordinary execution, using privileged teacher actions for training. This lets the hand respond to its current state without waiting for another high-level call.

We instantiate this design by adapting a ViserDex-style RGB-to-pose policy [2] to a continuous semantic goal input while reusing compatible recurrent weights. We call the complete REVEAL system with the Astra agent "Full Astra." This route emits public tool arguments that map to the shared goal bank; it does not expose hidden model embeddings. A fine-tuned Qwen3-VL baseline [7] predicts a goal vector directly, and a rule-based VLM parses a color and readability decision. These interfaces provide a common motor policy while retaining the differences between an agent with feedback and history and single-step grounding baselines.

Across 1,440 simulation trials spanning six object-specific systems and two semantic mappings, REVEAL attains object-averaged success rates of 70.4% on original mappings and 65.8% on changed mappings. These exceed the strongest evaluated baseline mean in each setting by 26.2 and 40.4 percentage points, respectively. The means describe complete systems with object-specific weights and perception, not zero-shot policy transfer. A separate 2,160-trial English/Chinese study favors REVEAL over fine-tuned grounding in all eight duck and hedgehog conditions after multiplicity correction; cube differences remain unresolved. Matched cube interventions separately test fresh visual evidence and local control.

Our contributions are threefold. First, we formulate language-guided target presentation as online visual search and semantic binding during continuous in-hand control. Second, we develop an asynchronous agent–motor interface with public semantic actions and teacher-supervised local adaptation. Third, we evaluate that interface through six-object mapping comparisons, three-object language coverage, and matched cube interventions that distinguish the effect of fresh visual evidence from unresolved controller gains.

Task mappings, checkpoints, stopping rules, and physical success criteria are fixed within each cohort. Service exclusions are applied symmetrically within comparisons, with all raw outcomes retained. We report physical reach separately from correct semantic binding and autonomous completion.

## 2. Related Work

**Visual dexterous manipulation.** ViserDex uses monocular visual pose estimation and recurrent control for dexterous reorientation, with Gaussian representations supporting visual training [2]. We reuse this perception-control decomposition and adapt a released Revo3 policy to semantic goals. Closely related, GoalDex conditions a visuomotor policy on current and goal images and uses a VLM to select goal images from language [1]. It studies image-goal reorientation and multi-object sim-to-real transfer. REVEAL instead addresses online semantic search when the requested marked face may be occluded: the agent obtains new evidence and binds the target during manipulation, while local control continues during the external request. GoalDex also models visual-processing delays; our distinction is this decision and control interface, not asynchronous computation itself. ManipTrans studies residual learning for dexterous bimanual transfer [5]. REVEAL uses local residual adaptation to support the semantic interaction loop.

**Language and visual agents.** Magma connects multimodal representations to agent actions [11]. VERA translates visual plans into robot actions and includes language-conditioned dexterous in-hand reorientation [6]. Its hand experiments include directional rotation prompts. Our setting focuses on identifying a requested marked face under occlusion, maintaining contact during external reasoning, and checking the bound canonical target at autonomous stopping. The fresh-image intervention directly tests the need for updated visual evidence in this setting.

**Asynchronous robot policies.** Real-time chunk execution [3], online corrections for VLA chunks [9], and latency-aware RL [12] address inference delays through different execution and learning designs. DEFLECT studies delay-robust asynchronous VLAs through temporal counterfactual preferences [13]. LiMA combines sparse semantic intention and asynchronous action generation for dexterous manipulation [4]. REVEAL learns compact teacher-supervised controllers around a separate motor policy and public semantic tools. Its evaluated distinction is the online observation–binding–control loop; asynchronous computation and residual actions alone are not novel. The present controller ablations do not establish an advantage from explicit goal conditioning.

## 3. Task and Evaluation Contract

**Semantic target.** Q1 covers six object-specific simulation releases: a marked cube, a wooden die, a duck, a hedgehog, a pig, and a milk carton. Q2/Q3 interventions use only the marked cube. Cube and marked-object instructions identify one of six surface letters; die instructions request one through six dots. The cube letters are $\{T, E, R, L, S, H\}$. Each target has a canonical world orientation $Q(g)$ that places its marked face upward. The environment scores the

full canonical quaternion, including its fixed twist, rather than accepting every orientation with the correct normal. The language prompt is therefore less specific than the scoring convention. All methods use the same scoring convention.

At step $t$, the agent observes RGB evidence $I_t$ and the instruction $\ell$. The motor system observes proprioception $x_t$ and measured pose $\hat{p}_t$. The cube obtains $\hat{p}_t$ from RGB; the other releases use a noisy simulator-pose model. Clean true pose $p_t$ and teacher actions are used for supervision and independent scoring, and future semantic responses are unavailable to online control. We denote the active goal vector by $g_t$, recurrent state by $h_t$, and the pending-request flag by $m_t$.

**Physical time.** One request may be pending at a time. The cube protocol allows six model calls per episode and 3,600 logical control steps at 30 Hz; the other objects use 20 Hz over the same 120-second logical horizon. If a call starts at $k$ and takes $L$ wall-clock seconds, its response becomes usable only after the call completes and at least $\lceil 30L \rceil$ control steps have elapsed. Physics continues throughout this interval. The simulator currently runs slower than real time. This protocol represents inference delay in simulated physical time, but does not demonstrate a 30 Hz hardware pipeline.

**Autonomous completion.** All cube systems share a measured-pose stopping gate, evaluated at 3 Hz. It requires a declared binding for the requested symbol, no pending request, a non-inspection state, and measured full-orientation error at most $\epsilon = 0.16$ radians. The gate may stop independently of an agent's native stop suggestion. The other object releases use calibrated completion correction for Astra and a measured-pose gate for bare baselines, with the same independently scored tolerance; their stopping conditions are documented in the supplement. At stop time $\tau$, the independent success variable is

$$S = \mathbf{1}\left[\hat{b}_\tau = b^*(\ell),\ d(q_\tau, Q(b^*)) \le \epsilon,\ \neg D_{0:\tau}\right], \quad (1)$$

where $\hat{b}_\tau$ is the declared face binding, $q_\tau$ is true orientation, $D_{0:\tau}$ indicates an earlier terminal drop, and $d(q, Q) = 2\arccos(|\langle q, Q\rangle|)$ for normalized quaternions. Merely passing through a target does not count. No extra hold duration is added after stopping. A pose false stop and a semantically wrong binding are failures even if the hand retains the object. The frozen drop rule is distance of at least 0.24 metres from the nominal object grasp position.

**Matched reporting.** Unless stated otherwise, each arm uses 40 raw trials from two seeds, with 20 trials per seed and matching initial physical states and scored physical targets. Instructions also match within the control and feedback studies. In the remapping study, the requested letter changes with the mapping to preserve the physical target. A service error or measured model latency above 120 seconds is recorded as a service exclusion. For a panel of methods $\mathcal{C}$, the eligible set removes the union of excluded trial identifiers across $\mathcal{C}$. We report both raw and eligible outcomes, never replace an excluded trial, and never mix denominators from different panels. Long service delays remain visible without being interpreted as controller failures.

## 4. Method

REVEAL connects semantic decisions to motor control through a shared goal bank (Fig. 2). We detail the cube implementation below; other object releases use their own history-conditioned students, noisy pose feedback and maintenance actors, as specified in the supplement.

### 4.1. Reusing a visual pose-conditioned student

A frozen CNN predicts the image coordinates and depth of the object center and eight keypoints from a $120 \times 120$ RGB crop. Calibration and rigid fitting recover the measured position and XYZW orientation, together with a fit residual. Cube trials use rendered object views with live hand occlusion; deployment receives no ground-truth pose.

The inherited student expects geometric goals. We replace its goal channels with a learned projection of $g_t \in \mathbb{R}^{768}$:

$$z_t = \tanh(W_g g_t + b_g) \in \mathbb{R}^{64}, \quad (2)$$

$$(u_t, h_{t+1}) = \pi_\theta([x_t, z_t], \hat{p}_t, h_t). \quad (3)$$

Here $x_t \in \mathbb{R}^{109}$ contains target-free proprioceptive features and history, while $u_t \in \mathbb{R}^{21}$ gives normalized joint-target proposals. Compatible encoders, recurrent weights, actor weights, and normalization statistics are initialized from the released policy. Neither target quaternion nor relative-goal quaternion is an input to the migrated student.

We fit the new goal interface and adapt the recurrent core with frozen-teacher actions in a DAgger-style procedure [8]. Teacher state and actions are training-only. The cube deployment uses the frozen RGB pose estimator; object-specific policy and perception differences for the other releases are documented in the supplement.

### 4.2. Semantic grounding and public actions

The cube uses six normalized goal vectors $B = \{g_1, \ldots, g_6\}$ from cached SigLIP2 text features [10]; each is bound to a color skill and canonical asset orientation.

**Semantic agent interface.** Given the requested symbol and current image, the hosted agent emits a validated public

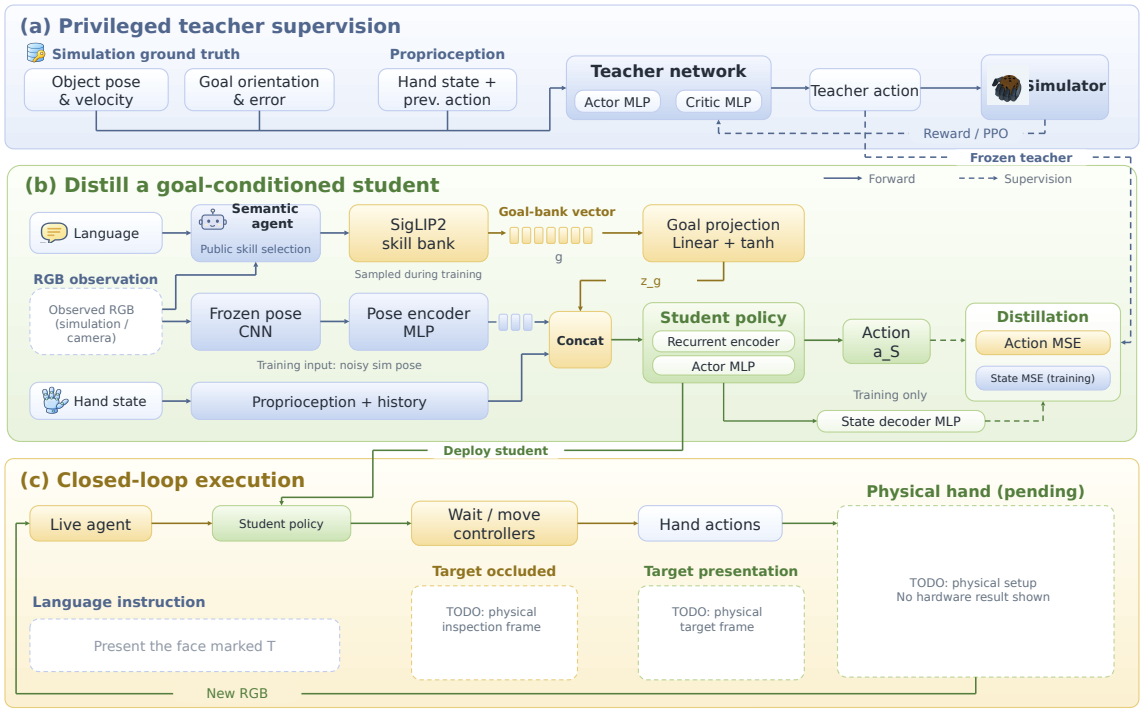



Figure 2. **REVEAL training and deployment pipeline.** Privileged supervision trains a student conditioned on a public goal-bank vector. The semantic agent selects skills, and local controllers adapt their execution. Dashed physical-hand slots are pending; they do not report hardware results.

action (inspect, reorient, verify, stop, or abstain). A color argument selects a vector from $B$; inspection may change the motor goal without binding the requested symbol. Agent-selected inspection/reorientation actions have bounded 10–90-step durations. Request latency is separate from action duration; updated images and public action history inform subsequent decisions. Public arguments map deterministically to the student goal; hidden agent features are not used.

### 4.3. Learning control while a request is pending

An outstanding query does not determine whether the current movement is still safe. We construct a deployable feature vector $s_t$ from proprioception, measured pose, student belief, proposed action, pose-fit residual, semantic binding status, and request status. It also includes the measured pose and command at the request anchor and elapsed pending time. The active goal $g_t$ is encoded separately. Before a binding is established it denotes the startup or inspection skill, not the unavailable future target.

A sensor encoder and a goal encoder feed a fusion layer. The waiting actor predicts a speed logit $r_t$ and a residual $v_t \in \mathbb{R}^{21}$. Let $\bar{q}_t$ be normalized current joint position and

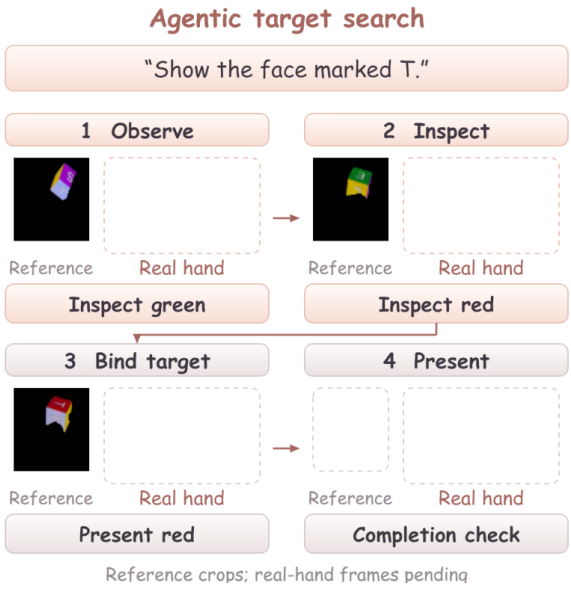



Figure 3. **Visual target search.** The agent observes, inspects candidate faces, binds the requested target, and issues a presentation command. The reference crops are simulated; real-hand frames are pending.

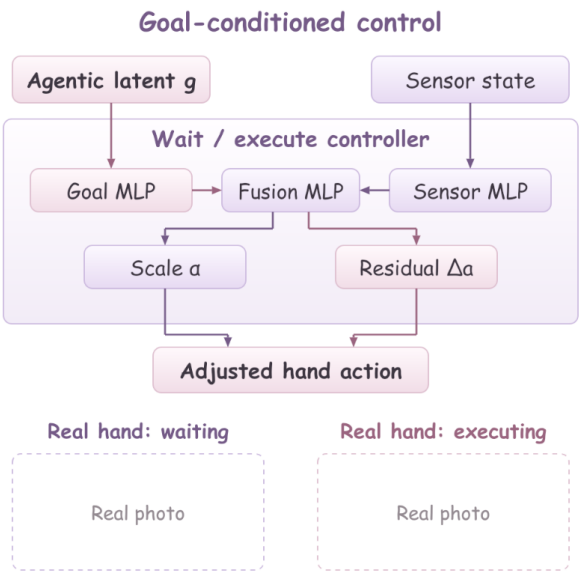



Figure 4. **Local control component.** A goal-conditioned actor combines sensor and goal features to scale the student proposal and add a bounded residual. The photo slots in the source illustration are pending and do not show real-hand results.

$\bar{u}_t$ the clipped student proposal. The applied command is

$$\alpha_t = \mathrm{sigmoid}(4r_t + \mathrm{logit}(0.15)), \tag{4}$$

$$b_t = \begin{cases} \bar{q}_t + \alpha_t(\bar{u}_t - \bar{q}_t), & m_t = 1, \\ \bar{u}_t, & m_t = 0, \end{cases} \tag{5}$$

$$a_t = \mathrm{clip}\big(b_t + 0.35\tanh(v_t), -1, 1\big). \tag{6}$$

Zero outputs reproduce fixed slowing with coefficient 0.15 while a query is pending. The environment's joint mapping and smoothing remain unchanged. The residual permits contact adjustments outside the interpolation between current joints and the student proposal.

Waiting targets are generated by the frozen teacher with the request anchor as the holding objective. This supervision is available in simulation. The deployed actor receives measurements and the active goal only. It never receives the answer being computed. The objective fits the applied action to the teacher label and penalizes residual magnitude:

$$\mathcal{L}_{\mathrm{control}} = \mathbb{E}_{(s,g,a^*)\sim\mathcal{D}}\left[\|a_\phi(s,g) - a^*\|_2^2 + 0.01\|\delta_\phi(s,g)\|_2^2\right]. \tag{7}$$

Refit protocol, checkpoint selection, and independent confirmation are reported in the supplement. The latest waiting-actor refit replaces only pending-request outputs; the student, predictor, execution controller, and stopping gate remain fixed.

### 4.4. Learning execution after a decision

When $m_t = 0$, a separate execution actor supplies $v_t$ in Eq. 6, using the student proposal as its base. It learns a bounded teacher-supervised correction for non-pending execution. Its active goal is explicit, although the goal-input interventions do not establish a benefit from this pathway (Sec. 5).

### 4.5. Closed-loop execution

At each control step, the system updates visual pose, advances the recurrent student, chooses the waiting or execution actor, and applies the resulting joint targets. A completed semantic response is validated before it changes the active skill or declared binding. The next image reflects the motion that occurred while the previous response was pending. The common stopping gate is checked separately from semantic action generation. Invalid model output cannot supply ground-truth information or silently select the requested target.

Student adaptation and controller learning are evaluated separately. A new student changes the belief and proposal distribution seen by the controllers. Their checkpoint-bound weights therefore require compatibility validation before transfer.

## 5. Experiments

We organize evaluation around four questions. Q1 tests complete systems, Q2 studies local control, and Q3 tests visual feedback. Q4 specifies the pending hardware evaluation. Q2 and Q3 use only the marked cube.

**Q1.** How do complete systems compare across objects, mappings and instruction variants?
**Q2.** How should the hand act during deliberation?
**Q3.** Does continued visual feedback help?
**Q4.** Does the system work on physical objects?

### 5.1. Setup and reporting

Experiments use Revo3 assets in Isaac Lab 3.0/Newton with validated contact capacities. Unless specified, each condition has two evaluation seeds and 40 scheduled trials. Physical tasks are paired within each study. Cube systems share the frozen visual pose estimator, goal bank, latent student, 0.16-radian tolerance and measured-pose stopping. Other objects use separate students and noisy simulator pose; Astra additionally uses calibrated completion correction. Bare VLM baselines omit the added control actors and use a fixed unreadable-view schedule, while Astra adapts its inspection. "Full Astra denotes the complete REVEAL deployment, not the semantic model alone. Natural model latency and these backend differences are part of the complete-system comparison.

The latest cube release replaces only its waiting actor; release cohorts and interventions remain separate. Main rates use common service-eligible slots across compared methods. Raw records are retained without replacement; Wilson intervals describe individual rates. Holm correction covers the waiting and visual contrasts within their respective families; confirmation and image refresh form a separate two-test family. Historical paired tests remain exploratory and unadjusted.

## 5.2. Q1: Systems across objects and mappings

We compare original and changed semantic mappings across six object-specific releases. Historical native-image cohorts remain separate in the supplement; their Astra–fine-tuned differences are unresolved (cube $p = 0.267$, die $p = 0.4545$).

**Aggregate system performance.** Over 1,440 scheduled trials, equal-object mean success is 70.4% on original mappings and 65.8% on changed mappings. The strongest baseline means are 44.2% (fine-tuning) and 25.4% (rules), yielding gains of 26.2 and 40.4 percentage points. These post-hoc descriptive means weight six service-eligible object rates equally; they neither pool trials nor add a significance test.

**Four additional objects.** Duck, hedgehog, pig and milk carton contribute 960 trials across two mappings. Astra has the highest observed rate in each cell (Table 1); these are object-specific systems, not common-policy transfer. Three historical service failures are excluded symmetrically. Milk-carton Astra drops in 15/40 original and 20/40 changed-mapping trials, so higher completion does not imply stable grasping. Reset and sensitivity audits are in the supplement.

**Visual-semantic intervention.** Only the upper image changes; predictor input and lighting remain fixed. Identity, cyclic letter–color permutation and masking use 40 matched states per method, all with reconstructed face images. Astra retains 32/40 successes after permutation; fine-tuning falls from 30/40 to 16/40 (Holm $p = 0.0130$, six contrasts); rules remain at 23/40. Masking yields 0/39, 3/40 and 0/40, respectively (full supplement table). Nine of eleven fine-tuned wrong bindings after permutation select the old association, consistent with learned shortcuts but not an exclusive memorization mechanism. One permutation does not establish broad generalization or equivalence.

**Dice layout transfer.** A separate 240-trial study rotates only the native visual mesh, preserving collision, inertia and the prompt distinguishing printed counts from motor-skill IDs. Table 1 reports 40 matched trials per cell, with no exclusions. Permuted-layout Astra contrasts give exploratory paired $p = 2.98 \times 10^{-8}$ against fine-tuning and $p = 5.96 \times 10^{-8}$ against rules; its layout contrast ($p = 0.688$) does not establish equivalence. Astra also receives probes, history and noisy feedback, so this supports system-level transfer, not isolated model superiority.

**Language and wording coverage.** Three object-specific cohorts test English/Chinese standard and rewritten instructions: three evaluation seeds, 60 scheduled trials per cell, 2,160 total. Table 2 uses common service-eligible slots; weights, mappings and scoring remain fixed, without Chinese retraining. The rule baseline also localizes its task/output instructions, so this tests deployed-system coverage. Rates are conditional on service availability; the supplement retains raw outcomes.

Astra reaches 67.8–76.7% on duck and 73.3–76.7% on hedgehog, exceeding fine-tuning in all eight conditions (Holm-adjusted paired $p \leq 0.0258$ across 36 method contrasts). Against rules, six of eight contrasts survive correction; English-standard differences remain unresolved. On cube, Astra leads numerically in three conditions, while fine-tuning leads on Chinese rewriting (81.1% versus 67.9%); none of their four contrasts is significant (all adjusted $p = 1.0$). Strict pairing sensitivity preserves these decisions. No language/wording contrast survives its separate 36-test correction; this does not establish language equivalence.

## 5.3. Q2: Controller ablations on the cube

On the marked cube, we compare four ways of acting while a semantic request is pending: continue the student proposal, interpolate toward it with fixed speed 0.15, learn that speed, and learn both speed and a bounded joint correction. Every condition uses the same frozen execution actor outside pending intervals. Astra, the student, visual pose predictor, stopping gate, and physical horizon are fixed.

The speed-only actor is trained with zero waiting residual, matching full-actor data, initialization, optimizer, sampling and 2,400 updates. Reproduction yields identical deployed full-actor tensors. Held-out checks verify identical non-pending actions.

Continued execution succeeds in 25/40 trials, fixed slowing in 30/40, learned slowing in 28/40, and learned slowing with joint correction in 34/40 (Table 3). There are no service exclusions. The full actor has the highest observed success, but learning only the speed does not improve on fixed slowing. Holm-adjusted exact paired $p$-values for the three registered adjacent contrasts are 0.604, 0.804, and 0.539. This cohort does not resolve those component gains.

Total drops are twelve, four, eight, and five. Full control has four later-pending drops versus one for learned slowing. Fixed slowing has four startup drops and no later drops.

| Object | Motor pose | $n_{\text{orig}}/n_{\text{changed}}$ | Original mapping | | | Changed mapping | | |
|---|---|---|---|---|---|---|---|---|
| | | | Rule | Fine-tuned | Full Astra | Rule | Fine-tuned | Full Astra |
| Cube | RGB pose | 40 / 40 | 57.5 | 75.0 | **80.0** | 57.5 | 40.0 | **80.0** |
| Die | Noisy pose | 40 / 40 | 32.5 | 57.5 | **70.0** | 2.5 | 0.0 | **65.0** |
| Duck | Noisy pose | 38 / 40 | 39.5 | 36.8 | **65.8** | 22.5 | 5.0 | **60.0** |
| Hedgehog | Noisy pose | 39 / 40 | 43.6 | 33.3 | **71.8** | 25.0 | 5.0 | **80.0** |
| Pig | Noisy pose | 40 / 40 | 52.5 | 40.0 | **92.5** | 37.5 | 7.5 | **82.5** |
| Milk carton | Noisy pose | 40 / 40 | 32.5 | 22.5 | **42.5** | 7.5 | 2.5 | **27.5** |
| Mean | Object-specific | – | 43.0 | 44.2 | **70.4** | 25.4 | 10.0 | **65.8** |

Table 1. **Q1: six objects and two semantic mappings.** Completion (%) uses common service-eligible trials; each cell has 40 scheduled trials. Mean weights six objects equally. Bold denotes the highest observed rate, not significance. Object-specific weights remain fixed across mappings; bare baselines omit added control modules. Cube uses reconstructed upper images and RGB pose; die rotates its visual mesh; other objects permute surface letters with noisy simulator pose. Native-image and language cohorts remain separate.

| | English | | Chinese | |
|---|---|---|---|---|
| System | Standard | Rewrite | Standard | Rewrite |
| **Cube** ($n$) | 56 | 59 | 57 | 53 |
| Rule VLM | 48.2 | 45.8 | 31.6 | 32.1 |
| Fine-tuned VLM | 71.4 | 72.9 | 75.4 | **81.1** |
| Full Astra | **75.0** | **79.7** | **84.2** | 67.9 |
| **Duck** ($n$) | 60 | 59 | 60 | 60 |
| Rule VLM | 50.0 | 25.4 | 33.3 | 31.7 |
| Fine-tuned VLM | 38.3 | 30.5 | 31.7 | 35.0 |
| Full Astra | **68.3** | **67.8** | **68.3** | **76.7** |
| **Hedgehog** ($n$) | 60 | 60 | 52 | 60 |
| Rule VLM | 46.7 | 38.3 | 36.5 | 36.7 |
| Fine-tuned VLM | 38.3 | 31.7 | 34.6 | 25.0 |
| Full Astra | **73.3** | **76.7** | **75.0** | **76.7** |

Table 2. **Q1: language and wording coverage.** Completion (%) on common service-eligible trials. Methods share the listed $n$ of 60 scheduled slots. Bold denotes the highest observed rate, not significance. Raw counts and paired analyses are in the supplement.

| Waiting policy | Success | % | Drops | Wait drops |
|---|---|---|---|---|
| Continue student | 25/40 | 62.5 | 12 | 7 |
| Fixed slowdown | 30/40 | 75.0 | 4 | 4 |
| Learned slowdown | 28/40 | 70.0 | 8 | 1 |
| Slowing + correction | 34/40 | 85.0 | 5 | 4 |

Table 3. **Q2: cube controller ablation.** Each condition has 40 scheduled trials. All arms use the same frozen controller outside pending requests. Successes and drops use the common service-eligible panel. Wait drops combine startup and later pending intervals. All 40 trials per arm are service-eligible.

Highest completion therefore does not coincide with fewest drops. The supplement separates phases. Shared execution weights do not imply identical states or subsequent agent decisions.

Trials use seeds 185501 and 185502 with reversed condition order for the second seed. Checkpoints are fixed before these outcomes. Historical training-version comparisons remain separate from this structural ablation.

**Independent confirmation.** Without refitting, a new two-seed cohort gives 25/39 successes for continuation, 31/39 for fixed slowing, and 30/39 for learned slowing with correction. One service slot is excluded symmetrically, with all 120 raw trials retained. Learning versus fixed slowing has six versus seven exclusive successes (Holm $p = 1.0$). The learned arm has no waiting drops but eight execution drops, compared with three and two for fixed slowing. This confirmation does not establish a learned-control advantage. Later failures may reflect states and commands induced by waiting, despite identical execution weights. The supplement reports the full panel and separately retains collaborator aggregates pending source-record reconciliation; those aggregates do not support the main controller claim.

**Other component evidence.** Earlier waiting–execution factorial tests do not establish a reliable interaction. Explicit controller-goal interventions also do not establish a benefit, and a separate learned execution-residual gate fails its development admission rule. These negative results and the matched-state diagnostic are retained in the supplement. The new waiting study changes neither the execution actor nor its goal input.

### 5.4. Q3: Visual feedback on the cube

On the same cube, we compare Astra under two observation protocols. One receives the initial RGB observation and measured feedback, then commits to a complete plan containing one to six bounded actions. It receives no later semantic observation. Its last non-abstain skill can continue until the shared stopping gate or horizon terminates the trial. The other can use up to six calls with updated observations and history. Both retain continuous pose feedback to the

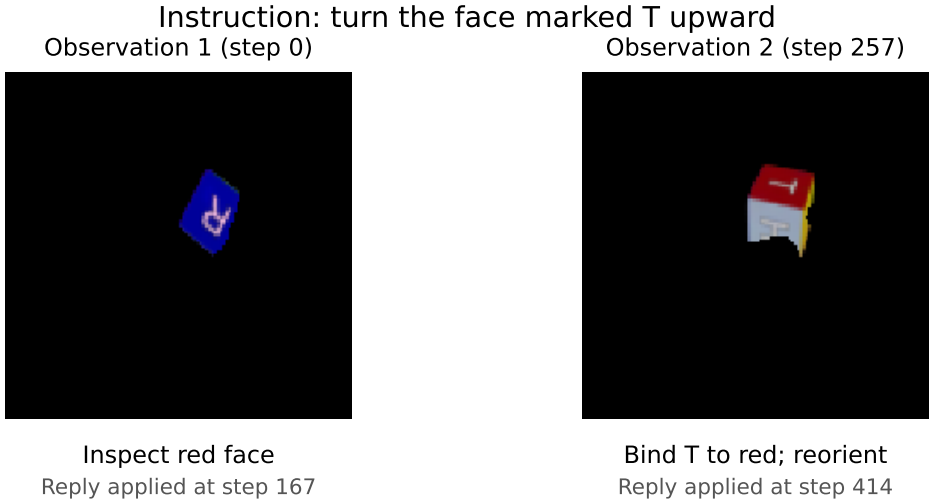



Figure 5. **Recorded inspection before binding.** Unretouched model-input frames from simulated trial 185501000. The initial plan abstains; feedback reveals T, then Astra binds T to red and completes at step 420 (true error 0.0812 radians). Both arms start from identical state and image. This selected example illustrates evidence acquisition, not wrong-binding correction or hardware performance.

| Protocol | Success | % | Drops | Calls |
|---|---|---|---|---|
| One initial plan | 6/39 | 15.4 | 0 | 40 |
| Full feedback | 33/39 | 84.6 | 5 | 120 |

Table 4. **Q3: initial plan versus feedback on the cube.** Each arm has 40 scheduled trials. The full-feedback condition is shared with Q2, rather than counted as another 40 raw trials. Calls include all raw episodes. One service-affected slot is excluded symmetrically. Different inference budgets are part of the protocol comparison.

| Upper RGB | Success | % | Drops | Calls |
|---|---|---|---|---|
| Current image | 32/40 | 80.0 | 3 | 112 |
| Repeated first frame | 9/40 | 22.5 | 4 | 98 |

Table 5. **Q3: fresh visual evidence on the cube.** Forty trials per arm, with the same call cap and frozen local controllers. Only upper-level RGB is frozen. Calls are realized totals, not matched compute.

student, the same full waiting and execution actors, and the same stopping rule.

The initial plan succeeds in 6/39 eligible trials, compared with 33/39 for continued feedback. Twenty-seven paired trials succeed only with feedback and none only with the initial plan (exact $p = 1.49 \times 10^{-8}$). Raw success is 6/40 versus 34/40. One initial-plan HTTP 503 timeout is excluded from both arms without replacement. Actual calls total 40 and 120.

The initial plan abstains in 28 eligible trials, of which 22 succeed with feedback. Among eligible feedback trials, 36 contain inspection, 32 first bind after inspection, and 31 successes use multiple calls. These traces support obtaining evidence before binding, without demonstrating correction of an executed wrong binding. The protocols differ in inference budget. Zero initial-plan drops accompany frequent abstention, not better completion. Figure 5 shows a recorded example; the supplement retains its selection rule and the earlier cohort.

**Does the next image matter?** A separate cohort retains the same six-call cap and trigger policy in both arms, changing only the upper agent's image. One arm receives current RGB, the other the exact first frame on every call. Numeric pose feedback, action history, lower visual perception and control remain live. Forty pairs share initial physical states and first images, with no service exclusions.

Current images yield 32/40 completions versus 9/40 for repeated first frames (Table 5). All 23 discordant pairs favor current images, a 57.5-point gain (exact $p = 2.38 \times 10^{-7}$, Holm $p = 4.77 \times 10^{-7}$). Abstentions fall from 24 to two, while drops change from four to three. Transport hashes verify the delivered images. This supports the value of later visual evidence with other feedback retained. Actual calls, tokens and trajectories differ, so equal maximum calls do not establish equal realized compute.

### 5.5. Q4: Multi-object hardware evaluation

**Hardware protocol.** **[TODO: Run and audit the multi-object physical-hand evaluation.]** The planned protocol compares the complete system and two baselines on each

available object. Report completion, drops, false stops, inference cost, calibration and synchronized videos. No physical-hand outcomes are available yet. The supplement specifies resets, independent scoring and video logging for this evaluation.

**Supporting cube perception study.** On 6,144 rendered observations, RGB-D reaches 64.73% pose recall versus 37.53% for RGB at 10 mm/10°. This independent estimator comparison uses ideal target masks and virtual depth; it does not establish downstream task gains or physical-camera accuracy. Full protocol and occlusion analysis are in the supplement.

## 6. Discussion and Limitations

**What the evidence establishes.** Fresh upper-level images improve cube completion with other feedback retained. Q1 does not isolate the semantic model from inspection schedules, control actors or stopping gates. Two languages and two templates do not establish broad multilingual generalization or equivalence.

**What remains unresolved.** Learned waiting control has the highest point estimate in the structural cohort, but independent confirmation gives 30/39 completions versus 31/39 for fixed slowing. Zero waiting drops coexist with eight execution drops. Explicit controller goals provide no demonstrated benefit; comparison with a generic time-aware residual controller remains open. Matched maximum calls do not imply equal realized compute, and Q2/Q3 mechanisms are tested only on the cube.

**Deployment scope.** Each object has separate weights. Cube pose comes from rendered RGB; other objects use noisy simulator pose. These experiments establish neither real-camera robustness nor zero-shot policy transfer. Only one changed mapping is tested per object. Full-quaternion scoring additionally fixes a twist not specified by the language request. Hosted-model revisions limit replication, and service-conditional rates do not measure end-to-end availability. Hardware evaluation remains pending.

## 7. Conclusion

REVEAL connects online visual search and semantic binding to continuous in-hand control. Across six object-specific simulation systems, mean success reaches 70.4% on original mappings and 65.8% on changed mappings. The language study extends coverage to English and Chinese, while matched cube interventions support the value of fresh images. Controller gains and physical-hand deployment require further validation.

## LLM Usage Statement

OpenAI Codex assisted with research discussion, implementation, experiment orchestration, literature retrieval, statistical analysis, figure-layout code, and drafting and editing throughout the manuscript. Astra and Qwen3-VL are evaluated system components. Numerical claims are grounded in recorded experiments; unfinished work is identified explicitly. The authors are responsible for verifying the text, references, code, figures, and results before submission.

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